

Uniform Schur bounds and holonomy in signed cycle squares

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Abstract

We give a direct Schur-complement proof of spectral bounds for explicit signings of cycle squares. A rationally certified four-dimensional Riccati recurrence produces a common six-dimensional limiting boundary matrix. Concatenating any number of prescribed cells, with unequal lengths and either Hamilton holonomy, preserves a squared spectral-radius bound of 7.92 when every cell has length at least 106, and 7.90537 when every cell has length at least 202. A degree-two estimate keeps the error independent of the number of cells. The one-cell family satisfies the sharper bound at every admissible size, and its supremum lies in an interval of width 10^{-6} . Exact finite bases complete explicit competitors to the twisted signing at orders 32, 40 and every even order at least 48. This recovers the previously known witness range through a common analytic mechanism. We also present the exact period-eight holonomy calculation and its application to the revised global-optimality conjecture. Unrestricted minima and minimizing signings are not determined.

1 Introduction and main results

For $N \geq 8$, the cycle square $C_N(1, 2)$ has vertex set $\mathbb{Z}/N\mathbb{Z}$ and edges $\{i, i+1\}$ and $\{i, i+2\}$. A real signing assigns ± 1 to each edge. For its symmetric adjacency matrix A , write

$$\rho(A) = \max\{|\lambda| : \lambda \in \text{spec}(A)\}, \quad \mu_N = \min_{\sigma} \rho(A_{\sigma}).$$

The minimization controls both spectral edges. The benchmark in the original twisted-optimality question [1] is

$$\rho_{\text{tw}}(N)^2 = 4 + 2 \cos \frac{2\pi}{N} + 2 \cos \frac{4\pi}{N} \quad (N \text{ even}, N \geq 8). \quad (1)$$

Our main result is a uniform bound for a family whose local structure is fixed but whose cell lengths and cell count are arbitrary. Set

$$t = (1, 1, -1, 1, -1, -1, 1, -1), \quad w_j = t^j \parallel (1, -1), \quad h_j = 8j + 2 \quad (j \geq 1). \quad (2)$$

Here $\parallel$ denotes concatenation. For a length- N triangle-sign word τ and $\alpha \in \{\pm 1\}$, define $A_N(\tau, \alpha)$ by the edge signs

$$a_i = \begin{cases} 1 & i < N-1, \\ \alpha & i = N-1, \end{cases} \quad \sigma\{i, i+1\} = a_i, \quad \sigma\{i, i+2\} = \tau_i a_i a_{i+1}, \quad (3)$$

with cyclic indices. The triangle signs are τ_i , and the Hamilton-cycle sign product is α . Define the two rational caps

$$c_0 = \frac{198}{25} = 7.92, \quad c_1 = \frac{790537}{100000} = 7.90537.$$

Theorem 1.1 (Unequal-cell bound). *Let $r \geq 1$, $j_0, \dots, j_{r-1} \geq 1$, and $\tau = w_{j_0} \parallel \dots \parallel w_{j_{r-1}}$, of total length N . For either holonomy α ,*

$$\min_i h_{j_i} \geq 106 \implies \rho(A_N(\tau, \alpha))^2 < c_0, \quad (4)$$

$$\min_i h_{j_i} \geq 202 \implies \rho(A_N(\tau, \alpha))^2 < c_1. \quad (5)$$

The constants do not depend on r , and the lengths need not be equal.

The allowed cell words are part of the hypotheses. The theorem does not cover arbitrary signings or arbitrary arrangements of local defects. Its proof retains six coordinates at each cell boundary and eliminates the four-site interior chains exactly. A common positive limiting core and a quadratic-form estimate with two chain incidences per boundary control the resulting $6r$ -dimensional cyclic matrix. The error is independent of r ; no count of localized eigenmodes is assumed.

The single-cell family permits a sharper finite completion.

Theorem 1.2 (One-cell bound). *For every $k \geq 1$,*

$$\rho(A_{8k+2}(w_k, +1))^2 < c_1.$$

Moreover,

$$\frac{7905369}{1000000} < \sup_{k \geq 1} \rho(A_{8k+2}(w_k, +1))^2 \leq \frac{790537}{100000}. \quad (6)$$

The interval in (6) has width 10^{-6} . Its upper endpoint is non-strict; no convergence, monotonicity, or exact supremum is asserted.

To cover each nonzero even residue modulo eight, put

$$W_{k,r} = \parallel_{a=0}^{r-1} w_{\lfloor (k+a)/r \rfloor}, \quad r \in \{1, 2, 3\}, \quad k \geq 6.$$

The cell parameters sum to k , so the word has length $8k + 2r$.

Theorem 1.3 (Explicit finite competitors). *For $r \in \{1, 2, 3\}$ and $k \geq 6$, the matrix*

$$B_{k,r} = A_{8k+2r}(W_{k,r}, (-1)^{r+1})$$

satisfies

$$\rho(B_{k,r})^2 < c_0 < \rho_{\text{tw}}(8k + 2r)^2. \quad (7)$$

For $r = 1$ the stronger cap c_1 holds. Together with the period-eight family, these constructions prove

$$\mu_N < \rho_{\text{tw}}(N) \quad \text{for } N \in \{32, 40\} \cup \{N \geq 48 : N \text{ even}\}.$$

This witness range already occurs in the earlier Target A project [6]. The contribution here is a common transparent analytic mechanism, sharper quantitative bounds, and a finite completion with directly specified signings. For odd k in residue four, and for $3 \nmid k$ in residue six, our placements need not be the older balanced-gap placements. Their finite certificates are recomputed for the words above. We do not reprove the universal lower bounds at complementary smaller orders, determine μ_N when the twisted signing fails, or classify minimizing signings.

The literature target has also changed. The original preprint [1] was withdrawn and merged into [2]. Its revised Theorem 26 supplies the positive-holonomy period-eight value

$$\eta_* = 4 + \sqrt{10 + 2\sqrt{5}}, \quad r_* = \sqrt{\eta_*}, \quad (8)$$

for $N = 8m$, $m \geq 4$, while Conjecture 28 proposes $\mu_{8m} = r_*$ throughout that range. The exact negative-holonomy formula from the earlier project manuscript [5] is strictly smaller at every finite size. We provide its full proof and apply it to that revised claim. This application is distinct from a priority claim for the inherited formula.

The use of holonomy belongs to the established character approach to spectral graph theory; Luo and Roy [3, Theorems 3.6 and 3.13] study character families and endpoint attainment. Those results do not supply the fixed-support minimum here. Likewise, signed moments averaged over all signings, as in Chen, van Dam, and Bu [4, Theorem 3.1], are not pointwise lower bounds for every signing. Our estimates instead arise from an explicit fiber calculation and exact positive-definiteness tests after a uniform analytic reduction. The latter proofs are finite-certificate-assisted, with independent exact replays; their finite checks are specified in Section 7.

2 Switching and finite holonomy

Write a_i for the sign of $\{i, i+1\}$ and d_i for the sign of $\{i, i+2\}$, with cyclic indices. The triangle sign and Hamilton-cycle holonomy are

$$\tau_i = a_i a_{i+1} d_i, \quad \alpha = \prod_{i=0}^{n-1} a_i.$$

Switching by vertex signs $g_i \in \{\pm 1\}$ replaces A by GAG , where $G = \text{diag}(g_i)$. It preserves the spectrum, every τ_i , and α .

Lemma 2.1. *For prescribed triangle signs $(\tau_i)_{i \in \mathbb{Z}/n\mathbb{Z}}$, there are exactly two switching classes, indexed by $\alpha \in \{\pm 1\}$. Each has a unique representative with*

$$a_0 = \cdots = a_{n-2} = 1, \quad a_{n-1} = \alpha, \quad d_i = \tau_i a_i a_{i+1}.$$

Proof. Set $g_0 = 1$ and recursively $g_{i+1} = g_i a_i$ for $0 \leq i < n-1$. After switching, the first $n-1$ step-one signs are positive. Their product with the remaining sign is α , so that remaining sign is α . The triangle equations force every step-two sign. Conversely, the displayed signs realize the prescribed data for either choice of α . A switch between two such representatives is constant along the path $0, 1, \dots, n-1$, hence constant on all vertices and acts trivially. Finally, distinct values of the switching invariant α cannot be switching equivalent. $\square$

For the repeated word t^m on $N = 8m$ vertices, write $A_m^\pm = A_{8m}(t^m, \pm 1)$. The positive representative has all step-one signs positive and step-two signs $t_{i \bmod 8}$. The negative representative reverses exactly the three distinct edges

$$\{N-1, 0\}, \quad \{N-2, 0\}, \quad \{N-1, 1\}. \quad (9)$$

Reversing only the Hamilton edge would change two triangle signs. The two additional step-two reversals are therefore essential.

Extend $u \in \mathbb{C}^{8m}$ to the integer line by

$$u_{i+8m} = \alpha u_i. \quad (10)$$

Let $t_i = t_{i \bmod 8}$. In either seam gauge the adjacency action is the restriction of

$$(Lu)_i = u_{i-1} + u_{i+1} + t_{i-2}u_{i-2} + t_i u_{i+2}. \quad (11)$$

In particular, the signs of all cut-crossing edges in the finite matrix agree with (10), including for $m = 1$.

Lemma 2.2. *The complexification of A_m^α is unitarily equivalent to*

$$\bigoplus_{z^m=\alpha} H(z),$$

where, in the order $0, 1, \dots, 7$,

$$H(z) = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & z^{-1} & z^{-1} \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & -z^{-1} \\ 1 & 1 & 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & -1 \\ z & 0 & 0 & 0 & -1 & 1 & 0 & 1 \\ z & -z & 0 & 0 & 0 & -1 & 1 & 0 \end{pmatrix}. \quad (12)$$

Proof. For each root $z^m = \alpha$ and $v \in \mathbb{C}^8$, set $u_{8j+a} = m^{-1/2} z^j v_a$ for $0 \leq j < m$ and $0 \leq a < 8$. This satisfies (10). Substitution into (11) gives (12). For two different allowed phases z, w , the ratio $z\bar{w}$ is a nontrivial m th root of unity, so

$$\sum_{j=0}^{m-1} (z\bar{w})^j = 0.$$

Thus the m fiber spaces are mutually orthogonal, each has dimension eight, and together they exhaust dimension $8m$. The displayed map is unitary. Since $|z| = 1$, the fiber matrix is Hermitian. $\square$

3 The exact antiperiodic spectrum

Let $A_m^\pm = A_{8m}(t^m, \pm 1)$ and set

$$R(s) = \sqrt{4 + \sqrt{8 + s + \sqrt{26 - 3s}}} \quad (-2 \leq s \leq 2).$$

Theorem 3.1. *For every $m \geq 1$,*

$$\rho(A_m^+)^2 = \eta_*, \quad \rho(A_m^-) = R\left(2 \cos \frac{\pi}{m}\right) < \sqrt{\eta_*}. \quad (13)$$

Among signings with the prescribed labeled triangle signs t^m , the negative-holonomy class is the unique minimizing switching class.

The exact formula is recorded in the earlier period-eight manuscript [5]; we include its proof to keep the present argument self-contained.

Lemma 3.2. *For $|z| = 1$ and $s = z + z^{-1}$,*

$$\det(xI_8 - H(z)) = P(x, s) := x^8 - 16x^6 + 80x^4 - 128x^2 + 38 + s(-2x^4 + 16x^2 - 13) + s^2. \quad (14)$$

Its eight roots, with all signs independently chosen, are

$$\pm \sqrt{4 \pm \sqrt{8 + s \pm \sqrt{26 - 3s}}}. \quad (15)$$

They are all real, nonzero, and distinct for every $-2 \leq s \leq 2$.

Proof. Expanding the determinant of (12) as a Laurent polynomial in z gives

$$x^8 - 16x^6 + 80x^4 - 128x^2 + 40 + (z + z^{-1})(-2x^4 + 16x^2 - 13) + (z^2 + z^{-2}).$$

Using $z^2 + z^{-2} = s^2 - 2$ proves (14). This is a finite polynomial identity and can also be checked by the exact symbolic determinant in the accompanying verifier.

Put $y = x^2$. A translation gives

$$P(x, s) = (y - 4)^4 - (16 + 2s)(y - 4)^2 + s^2 + 19s + 38. \quad (16)$$

Solving this as a quadratic in $(y - 4)^2$ gives

$$u_{\pm}(s) = 8 + s \pm \sqrt{26 - 3s}.$$

The function $s^2 + 19s + 38$ is increasing on $[-2, 2]$ and has value 4 at -2 . Since $8 + s > 0$, the identity

$$(8 + s)^2 - (26 - 3s) = s^2 + 19s + 38 \geq 4$$

proves $u_- > 0$. Also $u_+ > u_-$, and

$$u'_+(s) = 1 - \frac{3}{2\sqrt{26 - 3s}} > 0, \quad u_+(s) \leq 10 + 2\sqrt{5} < 16.$$

Consequently

$$0 < 4 - \sqrt{u_+} < 4 - \sqrt{u_-} < 4 + \sqrt{u_-} < 4 + \sqrt{u_+}.$$

These are the four distinct positive roots in y . Taking their positive and negative square roots gives exactly (15). $\square$

Proof of Theorem 3.1. Lemma 3.2 implies

$$\rho(H(z)) = R(s), \quad s = z + z^{-1}.$$

The derivative computation in that lemma and strict monotonicity of the outer square roots show that R increases strictly on $[-2, 2]$. For positive holonomy, the grid $z^m = 1$ contains $z = 1$, so its maximal value of s is 2. For negative holonomy, the grid is

$$z_j = \exp\left(\frac{(2j + 1)\pi i}{m}\right), \quad 0 \leq j < m,$$

and its maximal value of s is $2 \cos(\pi/m)$. Lemma 2.2 therefore proves both equalities in (13). For every finite $m \geq 1$, $\cos(\pi/m) < 1$, which proves the strict inequality. Lemma 2.1 shows that these are exactly the two switching classes for the prescribed triangle signs, so the strict comparison proves the stated constrained uniqueness. Since A_m^- is an admissible real signing, the same value is an upper bound for the unrestricted minimum. $\square$

To identify $\sqrt{\eta_*}$ with the algebraic number used in the revised conjecture, put

$$f(x) = x^4 - 2x^3 - 6x^2 + 12x - 4.$$

An exact expansion gives $P(x, 2) = f(x)f(-x)$. On $x \geq \sqrt{6}$,

$$f''(x) = 12(x^2 - x - 1) > 0, \quad f'(\sqrt{6}) = 12\sqrt{6} - 24 > 0, \quad f(\sqrt{6}) = -4.$$

Hence f has exactly one root on that ray. For $x > \sqrt{6}$, $f(-x) - f(x) = 4x(x^2 - 6) > 0$. The largest positive root of $P(x, 2)$ is $R(2) > \sqrt{6}$. It must be a root of f , because $f(-R(2)) = 0$ would force $f(R(2)) < 0$ and then a still larger positive root of f , contrary to maximality. Thus $R(2)$ is the largest real root of f .

Corollary 3.3. *The negative-holonomy radii increase strictly to r_* as m increases, and*

$$r_* - \rho(A_m^-) = \frac{\pi^2}{4r_*\sqrt{10+2\sqrt{5}}} \left(1 - \frac{3}{4\sqrt{5}}\right) m^{-2} + O(m^{-4}). \quad (17)$$

Proof. The sequence $2\cos(\pi/m)$ increases strictly to 2, and R is strictly increasing and smooth near 2. Since

$$2\cos(\pi/m) = 2 - \pi^2 m^{-2} + O(m^{-4}), \quad R'(2) = \frac{1 - 3/(4\sqrt{5})}{4r_*\sqrt{10+2\sqrt{5}}},$$

Taylor's theorem proves (17). $\square$

There is no parity exception: when m is odd, the allowed phase $z = -1$ has $s = -2$ and cannot recover the omitted maximum at $s = 2$. At $m = 1$ the single fiber is $H(-1)$, with $\rho(A_1^-) = \sqrt{6 + \sqrt{2}}$.

Corollary 3.4. *For every $m \geq 4$, $\mu_{8m} < \sqrt{\eta_*}$. Thus Conjecture 28 of [2] is false as stated.*

Proof. The matrix A_m^- is a real signing of the same graph, and Theorem 3.1 gives its strictly smaller radius. As shown above, $\sqrt{\eta_*}$ is the largest real root of the quartic used in that conjecture. The minimum in the conjecture ranges over both holonomies. $\square$

At $m = 4$, the exact value is

$$\rho(A_4^-) = \sqrt{4 + \sqrt{8 + \sqrt{2} + \sqrt{26 - 3\sqrt{2}}}} = 2.7842697993 \dots$$

A separate full-graph integer certificate checks that all leading principal minors of $279I_{32} \pm 100A_4^-$ are positive. Sylvester's criterion gives $\rho(A_4^-) < 2.79$. Exact evaluations

$$f(279/100) = -6766519/10^8 < 0, \quad f(14/5) = 76/625 > 0$$

and the monotonicity of f above show $2.79 < \sqrt{\eta_*} < 2.8$. This finite check is independent of the all- m fiber proof.

4 A common four-site chain and six-coordinate boundary

The remaining proofs use squared adjacency matrices. For a cap c , positive definiteness of $cI - A^2$ is equivalent to $\rho(A)^2 < c$. Throughout, $\|\cdot\|_2$ is the Euclidean operator norm and $\|\cdot\|_F$ is the Frobenius norm. We first derive one block system, then certify it at $c = c_0$ and $c = c_1$. No estimate at the second energy is inferred by simply reusing a certificate at the first.

For the one-cell word w_j , put $h = 8j + 2 = 4\ell + 2$, where $\ell = 2j$. Initially take positive holonomy and $h \geq 18$. Partition the vertices into

$$V_0 = \{0, 1\}, \quad V_s = \{2 + 4(s-1), \dots, 5 + 4(s-1)\} \quad (1 \leq s \leq \ell).$$

The normalized matrix $M_h(c) = cI - A_h(w_j, +1)^2$ has bulk diagonal and successive coupling blocks

$$D_c = \begin{pmatrix} c-4 & 0 & -1 & 0 \\ 0 & c-4 & 0 & -1 \\ -1 & 0 & c-4 & 0 \\ 0 & -1 & 0 & c-4 \end{pmatrix}, \quad (18)$$

$$E_+ = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 2 & 1 & 0 \\ 2 & -1 & 0 & -1 \end{pmatrix}, \quad E_- = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & -2 & 1 & 0 \\ -2 & -1 & 0 & -1 \end{pmatrix}. \quad (19)$$

Here $M_h[V_s, V_{s+1}] = E_+$ for odd s and E_- for even s . The boundary blocks are

$$\begin{aligned} G_0 &= (c-4)I_2, & H_0 &= D_c, & R_0 &= \begin{pmatrix} -1 & -2 & 1 & 0 \\ -2 & -1 & 0 & -1 \end{pmatrix}, \\ C_0 &= \begin{pmatrix} -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & -1 \end{pmatrix}, & W_0 &= \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \end{aligned} \quad (20)$$

Specifically, $R_0 = M_h[V_0, V_1]$, $C_0 = M_h[V_0, V_\ell]$, and $W_0 = M_h[V_1, V_\ell]$. All other nonadjacent bulk blocks vanish. These identities follow from the complete two-walk expansion

$$\begin{aligned} (A^2)_{i,i} &= 4, & (A^2)_{i,i+1} &= \tau_{i-1} + \tau_i, & (A^2)_{i,i+2} &= 1, \\ (A^2)_{i,i+3} &= \tau_i + \tau_{i+1}, & (A^2)_{i,i+4} &= \tau_i \tau_{i+2}, \end{aligned} \quad (21)$$

their transposes, and zero entries at all other cyclic offsets. At $h \geq 18$ these offsets and the stated block locations do not collide. The exceptional order $h = 10$ will be checked directly.

Use zero-based elimination indices: let $X_0 = D_c$, with $E_a = E_+$ for even a and $E_a = E_-$ for odd a . Eliminating the current four-site pivot gives the length-independent open recurrence

$$\begin{aligned} X_{a+1} &= D_c - E_a^T X_a^{-1} E_a, & R_{a+1} &= -R_a X_a^{-1} E_a, \\ W_{a+1} &= -E_a^T X_a^{-1} W_a, & G_{a+1} &= G_a - R_a X_a^{-1} R_a^T, \\ H_{a+1} &= H_a - W_a^T X_a^{-1} W_a, & C_{a+1} &= C_a - R_a X_a^{-1} W_a. \end{aligned} \quad (22)$$

The final pivot has index $p = \ell - 2$, which is even. At that step the right coupling is $W_p + E_+$, so the retained six-coordinate core is

$$S_\ell(c) = \begin{pmatrix} G_p - R_p X_p^{-1} R_p^T & C_p - R_p X_p^{-1} (W_p + E_+) \\ * & H_p - (W_p + E_+)^T X_p^{-1} (W_p + E_+) \end{pmatrix}. \quad (23)$$

The lower-left block is the transpose of the upper-right block. Repeated Schur congruence proves $M_h(c) \succ 0$ once every eliminated pivot and this core are positive. The subscript ℓ counts four-site blocks, not vertices.

4.1 The two finite rational certificates

Write $F_\pm(X) = D_c - E_\pm^T X^{-1} E_\pm$ and $\Phi = F_- \circ F_+$. At the even entrance J set

$$Z = X_J = \Phi^{J/2}(D_c), \quad Y = F_+(Z), \quad T_Z = Z^{-1} E_+ Y^{-1} E_-.$$

Both energies use the fixed rational weights

$$P = \frac{1}{10000} \begin{pmatrix} 10766 & 87 & 19 & 974 \\ 87 & 12664 & 148 & -2418 \\ 19 & 148 & 10093 & -25 \\ 974 & -2418 & -25 & 14009 \end{pmatrix},$$

$$Q = \frac{1}{10000} \begin{pmatrix} 11503 & 614 & 990 & -1101 \\ 614 & 10470 & 15 & 113 \\ 990 & 15 & 12299 & -2632 \\ -1101 & 113 & -2632 & 13260 \end{pmatrix}. \quad (24)$$

The numerical constants are exact rational numbers:

Quantity	$c = c_0$	$c = c_1$
Entrance J	24	48
Riccati radius δ	10^{-10}	10^{-18}
Response square bound ζ	10^{-10}	10^{-20}
Center transfer factor κ	$2/5$	$1/2$
Response contraction q	$2/3$	$3/4$
Riccati contraction $\theta = q^2$	$4/9$	$9/16$
Even response bound a	$1/30000$	$1/9000000000$
Pair response bound $b = 12a$	$1/2500$	$1/7500000000$
Seed margin γ	$1/50$	10^{-6}
Tail tolerance ϵ	$1/500$	10^{-8}
Seed graph order $4(J+2)+2$	106	202

Lemma 4.1 (Exact finite premises). *For each column of the table, the rational recurrence satisfies:*

1. $X_0, \dots, X_J \succ 0$ and $Z, Y \succ I_4/2$;
2. $(9/10)I_4 \prec P, Q \prec 2I_4$;
3. $T_Z^T P T_Z \prec \kappa P$ and $T_Z Q T_Z^T \prec \kappa Q$;
4. $\|\Phi(Z) - Z\|_F < \delta/40$;
5. $R_J Q R_J^T \prec \zeta I_2$ and $W_J^T Q W_J \prec \zeta I_4$;
6. $S_{J+2}(c) \succ \gamma I_6$.

Proof. All the matrices are specified by the rational initial data and at most $J+2$ recurrence steps. The accompanying exact verifiers evaluate them over $\mathbb{Q}$. For each positive-definiteness assertion they compute scalar Schur pivots: if a symmetric matrix is partitioned as $\begin{pmatrix} d & v^T \\ v & K \end{pmatrix}$, they require $d > 0$ and recurse on $K - vv^T/d$. Every pivot is a strictly positive rational number. The residual check is the exact comparison

$$\sum_{u,v} (\Phi(Z) - Z)_{uv}^2 < (\delta/40)^2.$$

The verifiers reconstruct the responses, and the certificates store the exact centers, seeds and positive pivots; the c_1 certificate also stores both entrance responses. An independent implementation constructs the full signed graph and uses scalar elimination, without generating the entrance through the four-site recurrence. It recovers the entrance after $4J$ scalar pivots,

subtracts the physical terminal E_+ coupling to isolate W_J , and checks the same inequalities. Both energies pass these rational checks. The implementation and certificate locations are given in Section 8. $\square$

This lemma is the finite computational part of the uniform argument. No floating eigenvalue or limiting-matrix approximation is used in its acceptance tests. The seed margins refer to the normalized six-coordinate cores in (23).

5 Uniform Riccati and boundary estimates

We give one analytic proof valid for either parameter column above. For a real symmetric matrix U , use the order-unit norm

$$|U|_P = \inf\{d \geq 0 : -dP \preceq U \preceq dP\} = \|P^{-1/2}UP^{-1/2}\|_2.$$

On the closed ball $\mathcal{K} = \{X = X^T : |X - Z|_P \leq \delta\}$, $P \prec 2I$ gives $\|X - Z\|_2 \leq 2\delta =: e$. The center bounds imply $X \succ I/3$, $\|X^{-1}\|_2 \leq 3$, and

$$\|X^{-1} - Z^{-1}\|_2 \leq 6e.$$

Since $\|E_\pm\|_F^2 = 14$, putting $Y_X = F_+(X)$ yields

$$\|Y_X - Y\|_2 \leq 84e < 1/6, \quad Y_X \succ I/3, \quad \|Y_X^{-1} - Y^{-1}\|_2 \leq 504e.$$

For $T(X) = X^{-1}E_+Y_X^{-1}E_-$, a two-term product difference gives

$$\|T(X) - T_Z\|_2 \leq 14(6 \cdot 3 + 2 \cdot 504)e = 14364e. \quad (25)$$

The norm conversion factor from the Euclidean norm to either

$$\|T\|_{P,\text{col}} = \|P^{1/2}TP^{-1/2}\|_2, \quad \|T\|_{Q,\text{row}} = \|Q^{-1/2}TQ^{1/2}\|_2$$

is at most $\sqrt{20/9} < 3/2$. Thus the perturbation in either norm is less than $43092\delta < 10^{-4}$. Both parameter columns satisfy $\sqrt{\kappa} + 10^{-4} < q$, and hence throughout the ball

$$T(X)^TPT(X) \prec q^2P, \quad T(X)QT(X)^T \prec q^2Q. \quad (26)$$

The derivative is $D\Phi_X[U] = T(X)^TUT(X)$. The first inequality in (26) therefore bounds its order-unit norm by $\theta|U|_P$, where $\theta = q^2$. Integration along a line segment in $\mathcal{K}$ proves the same Lipschitz bound for Φ . The center residual has order-unit norm below $\delta/36$. Both columns satisfy $1/36 + \theta < 1$, so

$$|\Phi(X) - Z|_P < \delta/36 + \theta\delta < \delta.$$

Banach's theorem gives a fixed point $X_* \in \mathcal{K}$. The actual even orbit starts at Z , whence

$$\|X_{J+2s} - X_*\|_2 \leq 4\delta\theta^s \quad (s \geq 0). \quad (27)$$

All these pivots and their intermediate odd partners exceed $I/3$. Together with Lemma 4.1, this proves positivity of every pivot at either energy.

5.1 Response decay and the limiting core

Two steps in (22) send R to $RT(X)$ and W to $T(X)^T W$. Their natural norms are

$$\|RQ^{1/2}\|_2, \quad \|Q^{1/2}W\|_2.$$

The second inequality in (26) contracts both by q . The table satisfies $a^2 > (10/9)\zeta$; the response premises and $Q \succ (9/10)I$ give

$$\|R_{J+2s}\|_2, \|W_{J+2s}\|_2 < aq^s. \quad (28)$$

A one-step transfer has norm at most $3\sqrt{14} < 12$. Consequently

$$\|R_a\|_2, \|W_a\|_2 \leq bq^s \quad \text{for } a \in \{J+2s, J+2s+1\}. \quad (29)$$

The orientation TQT^T is essential; a column-only estimate would not establish these two response bounds.

Define absolutely convergent series along the actual open trajectory:

$$\begin{aligned} G_\infty &= G_0 - \sum_{a \geq 0} R_a X_a^{-1} R_a^T, \\ H_\infty &= H_0 - \sum_{a \geq 0} W_a^T X_a^{-1} W_a, \\ C_\infty &= C_0 - \sum_{a \geq 0} R_a X_a^{-1} W_a. \end{aligned}$$

Convergence follows from (29) and $\|X_a^{-1}\|_2 \leq 3$ after the entrance. Set

$$S_\infty = \begin{pmatrix} G_\infty & C_\infty \\ C_\infty^T & H_\infty - E_+^T X_*^{-1} E_+ \end{pmatrix}. \quad (30)$$

There is no replacement of every pivot in the sums by X_* . Common initial terms therefore cancel exactly when a finite core is compared with this limit.

Proposition 5.1 (Complete one-cell tail). *For even $\ell \geq J+2$, write $p = \ell - 2 = J + 2s$. Then*

$$\begin{aligned} \|S_\ell(c) - S_\infty\|_2 &\leq B_s, \\ B_s &= \frac{24b^2q^{2s}}{1-q^2} + 48aq^s + 576\delta\theta^s < \epsilon. \end{aligned} \quad (31)$$

Furthermore $S_\infty \succ (\gamma - \epsilon)I_6$ and $S_\ell(c) \succ (\gamma - 2\epsilon)I_6$.

Proof. Each quadratic or mixed Schur suffix after p is bounded by

$$T_s = \frac{6b^2q^{2s}}{1-q^2}.$$

There are two single-block increments per pair, each at most $3b^2q^{2v}$; including the term at p only enlarges the bound. The terminal corrections in (23) have norms at most

$$\|R_p X_p^{-1} E_+\|_2 \leq 12aq^s, \quad \|W_p^T X_p^{-1} E_+ + E_+^T X_p^{-1} W_p\|_2 \leq 24aq^s.$$

The inverse identity and (27) give

$$\|X_p^{-1} - X_*^{-1}\|_2 \leq 36\delta\theta^s, \quad \|E_+^T (X_p^{-1} - X_*^{-1}) E_+\|_2 \leq 576\delta\theta^s.$$

Thus the errors in the G, C, H blocks are bounded respectively by T_s , $T_s + 12aq^s$, and $T_s + 24aq^s + 576\delta\theta^s$. Using

$$\left\| \begin{pmatrix} U & V \\ V^* & Z \end{pmatrix} \right\|_2 \leq \|U\|_2 + 2\|V\|_2 + \|Z\|_2$$

proves (31). Its right side decreases with s , and exact rational arithmetic gives

$$B_0 = \begin{cases} 251089/156250000 < 1/500, & c = c_0, \\ 583333407/10937500000000000 < 10^{-8}, & c = c_1. \end{cases} \quad (32)$$

Compare the certified seed $S_{J+2}(c) \succ \gamma I_6$ to the limit, and then the limit to any other finite core. These are two separate errors, giving the two stated lower bounds. $\square$

The resulting core margins are $2/125$ and $49/50000000$, respectively. Schur congruence transfers positivity to the full matrix; it does not transfer these numbers as Euclidean eigenvalue gaps of the full matrix.

5.2 Every unit complex phase

For $h = 8j + 2$ and $|z| = 1$, let $A_h(z)$ be the Hermitian cell operator induced by

$$(Au)_i = u_{i-1} + u_{i+1} + \tau_{i-2}u_{i-2} + \tau_i u_{i+2}, \quad u_{i+h} = zu_i, \quad \tau = w_j \text{ periodically extended.}$$

At $h \geq 18$, the two wrap blocks of $cI - A_h(z)^2$ are $C_0(z) = \bar{z}C_0$ and $W_0(z) = \bar{z}W_0$; all other initial blocks are unchanged. This follows directly from the two-walk paths, each wrap path crossing the cut once in the negative direction. The Hermitian Schur recurrence gives

$$X_p(z) = X_p, \quad R_p(z) = R_p, \quad G_p(z) = G_p, \quad H_p(z) = H_p, \quad C_p(z) = \bar{z}C_p, \quad W_p(z) = \bar{z}W_p.$$

Invariance of H_p uses conjugate transpose and $|z| = 1$. The physical terminal coupling remains $E = E_+$, so it is added to $\bar{z}W_p$, not multiplied by the phase. Put

$$\begin{aligned} g_p &= G_p - R_p X_p^{-1} R_p^T, & c_p &= C_p - R_p X_p^{-1} W_p, \\ h_p &= H_p - W_p^T X_p^{-1} W_p - E^T X_p^{-1} E, & u_p &= R_p X_p^{-1} E, \quad v_p = W_p^T X_p^{-1} E. \end{aligned}$$

The exact terminal core and its unitary conjugate are

$$\begin{aligned} S_\ell(z) &= \begin{pmatrix} g_p & \bar{z}c_p - u_p \\ * & h_p - zv_p - \bar{z}v_p^T \end{pmatrix}, \\ U_z^* S_\ell(z) U_z &= \begin{pmatrix} g_p & c_p - zu_p \\ * & h_p - zv_p - \bar{z}v_p^T \end{pmatrix}, \quad U_z = \text{diag}(I_2, zI_4). \end{aligned} \quad (33)$$

The lower-left blocks are conjugate transposes. The limit of (33) is the same S_∞ for every phase. Moreover, all the finite tail estimates in Proposition 5.1 are unchanged by the factors z , giving a uniform bound before taking any limit.

Corollary 5.2. *For every $|z| = 1$,*

$$h \geq 106 \implies \rho(A_h(z))^2 < c_0, \quad h \geq 202 \implies \rho(A_h(z))^2 < c_1,$$

where $h \equiv 2 \pmod{8}$ and the word is $w_{(h-2)/8}$. For any number of identical cells and either holonomy, the corresponding finite signing obeys the same applicable bound.

Proof. The uniformly positive pivots and phase core imply $cI - A_h(z)^2 \succ 0$. For r identical cells with total holonomy α , the maps

$$(F_z v)_{ah+s} = r^{-1/2} z^a v_s \quad (z^r = \alpha)$$

give an orthogonal decomposition of the full graph into the $A_h(z)$ fibers, by the same finite geometric-sum argument as Lemma 2.2. The seam identity is precisely $z^r = \alpha$. Every fiber is bounded, so the full radius is bounded. $\square$

6 Unequal cells and a bound independent of their number

The Fourier decomposition above requires identical cells. We now eliminate the interiors of unequal cells directly, retaining every boundary degree of freedom. This proves Theorem 1.1.

Let the cell lengths be $h_i = 8j_i + 2$, $0 \leq i < r$, and write $b_0 = 0$, $b_i = \sum_{a < i} h_a$, with all vertices interpreted modulo N . For each cell retain the ordered six-tuple

$$K_i = (b_i, b_i + 1, b_i - 4, b_i - 3, b_i - 2, b_i - 1) \quad (34)$$

and eliminate its interior

$$I_i = \{b_i + 2, \dots, b_i + h_i - 5\}.$$

Thus K_i contains the head pair of cell i and the tail four vertices of cell $i - 1$. The K_i and I_i partition the vertices for every legal $h_i \geq 10$.

The range-four expansion (21) shows that different interiors do not couple: adjacent interiors have closest separation seven. Distinct retained intervals also have no direct coupling, since their closest separation across cell i is $h_i - 5 \geq 5$. Each interior is a chain of $2j_i - 1$ four-site blocks; it couples only to K_i at its left and the tail-four slot of K_{i+1} at its right. The common initial and terminal triangle signs make these endpoint coefficients independent of the cell length.

In terms of (20), the intrinsic retained block and left and right chain couplings are

$$\mathcal{B} = \begin{pmatrix} G_0 & C_0 \\ C_0^T & D_c \end{pmatrix}, \quad U_0 = \begin{pmatrix} R_0 \\ W_0^T \end{pmatrix}, \quad V = \begin{pmatrix} 0_{4 \times 2} & E_+ \end{pmatrix}. \quad (35)$$

The global seam initially multiplies C_0 and the lower part of U_0 in K_0 by α . Conjugating the tail-four coordinates of K_0 by α restores (35) and moves the seam sign to the last chain's terminal coupling. Hence the canonical terminal factors are

$$\omega_i = 1 \quad (i < r - 1), \quad \omega_{r-1} = \alpha.$$

This is an explicit permutation and diagonal sign conjugacy.

6.1 The additive cyclic Schur identity

Use the same open pivot sequence as before and put

$$U_{a+1} = -U_a X_a^{-1} E_a.$$

Equation (22) gives $U_a = (R_a^T, W_a^T)^T$, that is, the vertical stack of R_a and W_a^T . For the final interior index $p_i = 2j_i - 2$, define

$$\mathcal{L}_i = \sum_{a=0}^{p_i} U_a X_a^{-1} U_a^T, \quad \mathcal{P}_i = V^T X_{p_i}^{-1} V, \quad F_i = U_{p_i} X_{p_i}^{-1} V. \quad (36)$$

Let $\iota_i : \mathbb{R}^6 \rightarrow \mathbb{R}^{6r}$ be the coordinate embedding at K_i , with the index $i + 1$ understood cyclically.

Lemma 6.1 (Exact cell assembly). *When the interior pivots are positive, their elimination gives the retained matrix*

$$\begin{aligned} \mathcal{S} = I_r \otimes \mathcal{B} - \sum_i \iota_i \mathcal{L}_i \iota_i^T - \sum_i \iota_{i+1} \mathcal{P}_i \iota_{i+1}^T \\ - \sum_i \omega_i (\iota_i F_i \iota_{i+1}^T + \iota_{i+1} F_i^T \iota_i^T). \end{aligned} \quad (37)$$

Every contribution is added, including coincident block locations when $r = 1$ or $r = 2$.

Proof. At the current pivot of a chain, the left coupling is U_a , so the Schur complement subtracts $U_a X_a^{-1} U_a^T$ from its retained left endpoint and propagates the response as $-U_a X_a^{-1} E_a$. Only the final pivot also sees the right endpoint, through $\omega_i V$. Its two self-energy terms and cross term are precisely (36). Distinct interiors do not couple, so their Schur contributions add. For $r = 2$, two different chains connect the same pair of retained cores; both cross contributions remain in (37). For $r = 1$, the combined coupling at the final pivot is $U_{p_0} + \omega_0 V^T$. Expanding its quadratic Schur correction gives $\mathcal{P}_0$ and the loop contribution $\omega_0(F_0 + F_0^T)$ in addition to $\mathcal{L}_0$. Thus the same additive formula also covers the one-cell case. $\square$

6.2 The common limit and the degree-two estimate

At either certified cap, suppose $p_i = J + 2s_i$ with $s_i \geq 0$, and put $s = \min_i s_i$. Define

$$\mathcal{L}_\infty = \sum_{a \geq 0} U_a X_a^{-1} U_a^T, \quad \mathcal{P}_\infty = V^T X_*^{-1} V.$$

By the stacked form of U_a , the matrix $\mathcal{B} - \mathcal{L}_\infty - \mathcal{P}_\infty$ equals exactly S_∞ from (30). Its positivity has already been proved; there is no new limiting-core assumption.

The separate response bounds give

$$\|U_{J+2v}\|_2 \leq \sqrt{2} a q^v, \quad \|U_{J+2v+1}\|_2 \leq \sqrt{2} b q^v.$$

Counting both increments in every pair therefore yields

$$\|\mathcal{L}_\infty - \mathcal{L}_i\|_2 \leq \frac{12b^2 q^{2s_i}}{1 - q^2}. \quad (38)$$

The right side harmlessly includes the final included term as well as the actual suffix. The inverse identity and $\|V\|_2 \leq \sqrt{14}$ give

$$\|\mathcal{P}_i - \mathcal{P}_\infty\|_2 \leq 576 \delta \theta^{s_i}. \quad (39)$$

Finally,

$$\|F_i\|_2 \leq 3\sqrt{28} a q^{s_i} < 16 a q^{s_i}. \quad (40)$$

The diagonal part of (37), relative to $I_r \otimes S_\infty$, combines one outgoing error from (38) with one incoming error from (39). For the cross terms, take arbitrary $v_i \in \mathbb{R}^6$. Their quadratic form has absolute value at most

$$\begin{aligned} \sum_i 2 \|F_i\|_2 \|v_i\|_2 \|v_{i+1}\|_2 &\leq \sum_i \|F_i\|_2 (\|v_i\|_2^2 + \|v_{i+1}\|_2^2) \\ &\leq 2 \max_i \|F_i\|_2 \sum_i \|v_i\|_2^2. \end{aligned}$$

Each retained core has exactly two chain-end incidences, counting a loop twice and retaining both parallel contributions. This proves the bound for every $r \geq 1$, without summing an error r times. We obtain

$$\|\mathcal{S} - I_r \otimes S_\infty\|_2 \leq E_s := \frac{12b^2 q^{2s}}{1 - q^2} + 576 \delta \theta^s + 32 a q^s. \quad (41)$$

Proof of Theorem 1.1. Exact arithmetic with the two parameter columns gives

$$E_0 = \begin{cases} 501647/468750000 < 1/500, & c = c_0, \\ 700000123/196875000000000000 < 10^{-8}, & c = c_1. \end{cases}$$

All terms in E_s decrease with s . Proposition 5.1 already supplies $S_\infty \succ (\gamma - \epsilon)I_6$, so (41) gives

$$\mathcal{S} \succ (\gamma - 2\epsilon)I_{6r} \succ 0.$$

The condition $p_i = 2j_i - 2 \geq J$ is exactly $h_i \geq 4(J + 2) + 2$, namely 106 or 202. All eliminated pivots are positive by Section 5. Schur congruence now proves $cI - A_N(\tau, \alpha)^2 \succ 0$, as required. $\square$

For each cell, the positive local quadrilateral fluxes $Q_u = \tau_u \tau_{u+1}$ occur at positions $0, 4, \dots, 8j - 4$ relative to its start. Successive positions have separation four within a cell and six across a cell boundary. Thus the construction has r gaps of length six and all other gaps of length four. The proof uses the legal cell words, not merely the number of such gaps. It is a full boundary-matrix argument and does not depend on the historical localized-mode count, physical interface-edge estimate, or IMS localization proof.

7 Finite completion and explicit competitors

The preceding analysis reduces each remaining statement to an explicitly bounded set of rational positive-definiteness tests. We give the complete ranges and the separating inequalities, including the cases where a stronger proposed cap is false.

7.1 The one-cell family and its uniform ceiling

Proof of Theorem 1.2. For $k \geq 25$, the single legal cell has length $8k + 2 \geq 202$, so Theorem 1.1 at $r = 1$, $\alpha = +1$, and $c = c_1$ applies. The remaining $k = 1, \dots, 24$ are exactly the orders

$$10, 18, 26, \dots, 194.$$

For each full signed graph the rational verifier proves $c_1 I - A_{8k+2}(w_k, +1)^2 \succ 0$ by positive scalar Schur pivots. The 10×10 matrix is constructed and checked directly, without assuming disjoint ordinary and wrap block locations. The first analytic order is 202, so no admissible size is missing.

At $c_- = 7905369/1000000$ and order 202, independent exact elimination of $c_- I - A_{202}(w_{25}, +1)^2$ has 196 positive scalar interior pivots. The retained six-coordinate core then has a strictly negative pivot after a positive scalar-pivot prefix. The recorded rational pivot is strictly negative, not merely nonpositive or a failed numerical test. Schur congruence gives a vector with negative quadratic form, and hence

$$\rho(A_{202}(w_{25}, +1))^2 > c_-.$$

This member supplies the lower endpoint in (6); the uniform strict upper bounds supply the non-strict upper endpoint. $\square$

7.2 Two and three nearly equal legal cells

For $r = 2$ and $k \geq 6$, the parameters $\lfloor k/2 \rfloor, \lceil k/2 \rceil$ sum to k . When $k \geq 26$, each is at least 13, so the cell bound c_0 applies. Exactly twenty finite cases remain:

$$k = 6, \dots, 25, \quad N = 52, 60, \dots, 204. \quad (42)$$

All use negative holonomy. For even k their cell lengths are equal. For odd k the two lengths are $N/2 - 4$ and $N/2 + 4$, which changes the placement from the older equally spaced gap construction. Each matrix in (42) is rebuilt from its own word and passes exact rational LDL positivity at cap c_0 . The first analytic order is 212.

For $r = 3$, write $k = 3q + s$, $0 \leq s < 3$. The three parameters

$$\left\lfloor \frac{k}{3} \right\rfloor, \left\lfloor \frac{k+1}{3} \right\rfloor, \left\lfloor \frac{k+2}{3} \right\rfloor$$

are $3 - s$ copies of q followed by s copies of $q + 1$. They sum to k , are at least two for $k \geq 6$, and differ by at most one. For $k \geq 39$ they are all at least thirteen, so Theorem 1.1 applies with positive holonomy. The remaining thirty-three cases are

$$k = 6, \dots, 38, \quad N = 54, 62, \dots, 310. \quad (43)$$

Each full signed graph passes exact rational LDL at cap c_0 . The first analytic order is 318, immediately after the final admissible finite order. When $3 \nmid k$, the cells are nearly equal rather than identical, and no certificate for another gap placement is substituted.

For both lists, the verification forms $(198/25)I - A^2$ or its denominator-cleared equivalent $198I - 25A^2$ directly from integer two-edge walks. A separate implementation uses the normalized matrix and a different scalar elimination order. For the three-cell list, it also reconstructs the triangle signs independently from the quadrilateral-gap word. Every required pivot in each list is strictly positive. These are the only finite ranges left by the minimum-cell-length reduction for the stated residue-four and residue-six families.

7.3 The twisted comparison and the all-even witness range

Proof of Theorem 1.3. For residue two, Theorem 1.2 gives the stronger cap $c_1 < c_0$ at every $k \geq 6$. The two- and three-cell arguments above give the cap c_0 for the other two residues. For $x > 0$, $\cos x > 1 - x^2/2$, and $\pi^2 < 10$. Therefore

$$\rho_{\text{tw}}(N)^2 > 8 - \frac{20\pi^2}{N^2} > 8 - \frac{200}{N^2}. \quad (44)$$

At $N \geq 50$, the last expression is at least $198/25$. Thus every nonzero residue witness satisfies (7). In particular, the exact endpoint gaps for the other two residues are

$$\frac{2679}{338} - \frac{198}{25} = \frac{51}{8450} > 0, \quad \frac{5782}{729} - \frac{198}{25} = \frac{208}{18225} > 0,$$

coming from $N = 52$ and $N = 54$ respectively.

For $N = 8m \geq 32$, Theorem 3.1 gives a witness with squared radius at most η_* . The rational comparison

$$\eta_* < \frac{999}{128} = 8 - \frac{200}{32^2} \quad (45)$$

follows by squaring positive quantities: the intermediate bound is

$$\frac{(999/128 - 4)^2 - 10}{2} = \frac{73329}{32768} > \sqrt{5}, \quad 73329^2 - 5 \cdot 32768^2 = 8433121 > 0.$$

Equations (44) and (45) prove the strict comparison at every such order. The four residue classes now cover 32, 40 and every even $N \geq 48$. $\square$

7.4 A stronger balanced subsequence and exact obstructions

Equal cells admit one additional small-order completion.

Proposition 7.1. *For every $j \geq 1$,*

$$\rho(A_{16j+4}(w_j \| w_j, -1))^2 < c_1.$$

For $j \geq 3$ this value is strictly below the twisted benchmark.

Proof. Corollary 5.2 applies for $j \geq 25$, with the two finite phases $z = \pm i$. The remaining $j = 1, \dots, 24$, at orders $20, 36, \dots, 388$, have exact full-graph positive-definiteness certificates for $790537I - 100000A^2$. Both primary and independent rational replays cover the complete list. Their smallest case uses a cell of length ten and is checked directly, including the coincident couplings. The first analytic order is 404. For $j \geq 3$, $N \geq 52$, so (44) and $c_1 < c_0$ give the comparison. $\square$

The sharper cap cannot replace c_0 in the full residue-four theorem. For the negative-holonomy nearly balanced words, exact rational elimination gives

$$\begin{aligned} \rho(A_{60}(w_3 \| w_4, -1))^2 &> c_1, \\ \rho(A_{76}(w_4 \| w_5, -1))^2 &> c_1. \end{aligned}$$

In each case the certificate records a strictly negative pivot with its complete positive predecessor prefix, proving a negative direction for $c_1I - A^2$. Thus the obstruction is exact and applies to the actual changed construction used here. There is also a distinct obstruction for the older balanced $N = 60$ gap word $[6, 4^6, 6, 4^6]$ with negative holonomy. Its half-length is thirty; its triangle signs change sign under that half-shift rather than repeating a legal w_j . It therefore lies outside Proposition 7.1, and its own exact negative pivot confirms that dropping the even- k condition would be false. None of these obstructions contradicts the c_0 bound or gives an unrestricted lower bound over all signings.

8 Verification, provenance, and remaining questions

The proof has three exact computational inputs: the fixed rational premises of Lemma 4.1, the explicitly bounded finite graph lists, and the strict lower obstructions. Everything that extends those data to unbounded lengths, phases, or numbers of cells is proved in Sections 5 and 6. The finite lists are summarized below; a separate seed check supplies each long-cell argument.

Family	Finite parameter range	Graph orders	Analytic from
One cell (c_1)	$1 \leq k \leq 24$	$10, 18, \dots, 194$	202
Two equal cells (c_1)	$1 \leq j \leq 24$	$20, 36, \dots, 388$	404
Two near-equal cells (c_0)	$6 \leq k \leq 25$	$52, 60, \dots, 204$	212
Three near-equal cells (c_0)	$6 \leq k \leq 38$	$54, 62, \dots, 310$	318

All acceptance comparisons use integers or rational numbers. Positive pivots certify positive definiteness by exact Schur congruence; negative pivots with positive predecessors certify the claimed strict obstructions. The scripts store construction parameters, pivot counts, exact certificate matrices or pivot data, and integrity digests. Digests identify outputs; the mathematical tests are the rational computations that generate them. Floating spectra used during exploration are excluded from the proof.

Independent replays reconstruct the underlying graphs or isolated chain responses separately, use different elimination orderings where applicable, and match the exact statements rather than trusting stored positivity flags. For example, the unequal-cell replay compares direct full-graph elimination with independently eliminated chains embedded additively into the cyclic boundary space, including loops and parallel contributions. These finite comparisons test the implementations; Lemma 6.1 proves the identity for every admissible length and number of cells.

The public supplement preserves the directories `certificates/analytic/`, `certificates/strengthening/`, `certificates/r4_pilot/`, `certificates/unequal_cells/`, and `certificates/r6_completion/`. Their respective primary verifiers are `verify_r2_certificate.py`, `verify_uniform_cap.py`, `verify_r4_pilot.py`, `verify_unequal_cells.py`, and `verify_r6_completion.py`. Each directory supplies its exact output. The first package’s independent replays are in `certificates/audit/`; the later packages have their own `audit/` subdirectories. `assembly.py` accompanies the unequal-cell verifier. The package README gives commands and required dependencies.

A separate Lean development verifies the exact negative-holonomy finite radius in Theorem 3.1. For the explicit raw graph matrix and every positive cell count, it proves that all Hermitian eigenvalue moduli are at most the stated radical and that an eigenvalue equals its positive value. The proof includes the seam and raw-operator bridges, maximal antiperiodic phase, determinant root, and lower attainment. The verified Lean 4.33.1 checkpoint audits 237 declarations, with axiom union exactly `propext`, `Classical.choice`, and `Quot.sound`. The separate quartic identification used to name the conjectured constant, the finite-size asymptotic, the integer principal-minor certificate, and the Riccati/Schur and residue-completion arguments are not formalized by that checkpoint. They retain their stated analytic and exact-certificate evidence.

The exact period-eight formula and the witness range are inherited project results; the present application to the revised conjecture and the common unequal-cell proof have their scope distinguished in the supplement’s claim ledger. The new proof requires neither the older physical interface-edge calculation nor an IMS localization estimate or a count of localized modes. It does not change the status of historical equality certificates at smaller orders.

The principal extremal problem remains to determine μ_N and its minimizing switching classes. Even within the prescribed one-cell family, the interval (6) does not determine the exact supremum or prove convergence. For broader cell arrangements, the two-cell obstructions at orders 60 and 76 show that shortening cells can affect the sharper cap. A structural lower bound over arbitrary triangle-sign words would be needed to turn any of these explicit upper bounds into a global minimization theorem.

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